



Semester II Examinations 2009/ 2010

Exam Code(s)	1MF1, 3BA1, 3BS9, 3EL2, 4BS2, 4CS2
Exam(s)	3 rd & 4 th Science, M.Sc. Software Design & Development

Module Code(s)	MP307
Module(s)	Modelling II

Paper No.	1
Repeat Paper	Yes

External Examiner(s)	Professor D.G.C. McKeon
Internal Examiner(s)	Dr P.T. Piirainen Dr M. Tuite

<u>Instructions:</u>	For full marks answer question 1 in Section A and TWO questions in Section B correctly.
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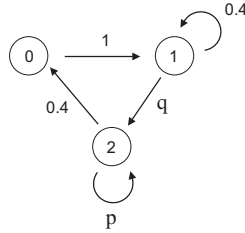
<i>Duration</i>	2hrs
No. of Pages	4
Department(s)	Applied Mathematics
Course Co-ordinator(s)	Dr P.T. Piirainen

Requirements:

MCQ	
Handout	
Statistical/ Log Tables	Yes
Cambridge Tables	
Graph Paper	
Log Graph Paper	
Other Materials	

Section A - Answer all questions

1. (a) Customer arrivals in a queue are described by a Poisson process with parameter α . Show that the probability for k customers to arrive in time t is given by the Poisson distribution with parameter αt . Hence find the average number of arrivals per unit time.
- (b) Consider the following Markov Process with transition probabilities shown.



- (i) Find the transition matrix P and the probabilities p and q .
 - (ii) Is the Markov process ergodic?
 - (iii) Find the equilibrium probabilities.
- (c) Consider the nonlinear population model

$$P_{n+1} = R(P_n)P_n$$

where

$$R(P) = \frac{rP}{1 + \left(\frac{P}{K}\right)^2}, \quad r, K > 0.$$

Find the fixed points for this model and show for what initial populations P_0 the population will stabilize and when it will die out .

- (d) Consider the linear control system

$$\begin{aligned} \frac{dx_1}{dt} &= x_2 + u, \\ \frac{dx_2}{dt} &= x_1 + x_2, \\ y &= x_1, \end{aligned}$$

where x_1 and x_2 represent the state, $u \in \mathbb{R}$ is the input and $y \in \mathbb{R}$ is the output.

- (i) Is this system observable?
- (ii) Is this system reachable?

Section B - Answer any **two** questions.

2. A queue of maximum size r has a Poisson arrival pattern with parameter α_k and Poisson servicing pattern with parameter β_k , where k is the queue size with $0 \leq k \leq r$.

- (a) Explain why $\alpha_r = \beta_0 = 0$. Assuming the system is ergodic show that the equilibrium probabilities are for $0 < k \leq r$ given by

$$\pi_k = \rho_k \rho_{k-1} \cdots \rho_1 \pi_0$$

where $\rho_k = \alpha_{k-1} / \beta_k$.

- (b) Consider a telephone system exchange with r operators. Assuming the probability of a call arriving is independent of the queue size, find the equilibrium probabilities.
- (c) An exchange consists of two operators, each of which can deal with 1 customer enquiry per 2 minutes on average. If 60 calls per hour arrive on average what is the equilibrium probability for all lines to be busy? What would be the effect of introducing a third operator?
3. Consider an ecological system with a prey population of size $x(t)$ and a predator population of size $y(t)$, which satisfy the differential equations

$$\begin{aligned} \frac{dx}{dt} &= (1 - 2y)x, \\ \frac{dy}{dt} &= (-1 + 2x)y. \end{aligned}$$

- (a) Give an interpretation for the various terms in these equations.
- (b) Show that the equilibrium points are $(x, y) = (0, 0)$ and $(x, y) = (1/2, 1/2)$. For small deviations away from the equilibrium point $(x, y) = (1/2, 1/2)$, show that the system is periodic with period 4π .
- (c) Show that in general the solution satisfies

$$K = \frac{e^{2(x+y)}}{xy}$$

for some constant K . Sketch a plot of x versus y and comment on the typical behaviour of this system in an application of this model.

4. A two-compartment model for drug delivery can be modelled as

$$\begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} -p_0 - p_1 & p_1 \\ p_2 & -p_2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} b_0 \\ 0 \end{pmatrix} u, \quad y = c_1,$$

where c_1 and c_2 are the concentrations of the drug in each compartment, p_0 , p_1 , p_2 and b_0 are rate constants, u is the flow rate of the drug into compartment 1 and y is the concentration of the drug in compartment 1.

- (a) Draw a schematic diagram of this system including concentrations, flow directions and rate parameters.
- (b) Investigate for what values of p_0, p_1 and p_2 this system is reachable.
- (c) Design a control function

$$u = -Kc + k_r r, \quad \text{for } K = \begin{pmatrix} k_1 & k_2 \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

where k_1, k_2, k_r are the control parameters, such that the desired output $r = 2$ and the closed loop eigenvalues for the system satisfies $\lambda_1 = -1$ and $\lambda_2 = -1$.